

[A101 - Toward Modifications of Extreme Weather Events](#)

Title: Weather Jiu Jitsu: Toward Adaptive Control of Extreme Weather Regimes Using the Seasonally Forced Lorenz-84 Model

Abstract:

As climate change intensifies extreme weather events, there is an urgent need for proactive, energy-efficient interventions to control catastrophic weather extremes. Weather Jiu Jitsu proposes to do this by leveraging the energy of the atmospheric circulation aided appropriately timed and located nudges. Our previous work on adaptive chaos control of idealized Lorenz 63 (L63) and Lorenz 84 (L84) models provided an initial proof of concept of the idea using idealized models. Here, we extend the concept of “Weather Jiu-Jitsu” to a L84 model whose two external forcings: equator to pole temperature gradient and land-ocean temperature contrast are allowed to vary seasonally. The L84 represents essential features of mid-latitude jet stream and eddy interactions and demonstrates intransitivity (weather regimes) and chaos (sensitivity to initial conditions). In past work we showed that its solutions are path dependent, e.g., fall and spring have the same external forcing in the model but given that they follow summer and winter, the outcomes as to the weather regime statistics are very different, as they are in the real world.

Using adaptive chaos control, we demonstrate that it is possible to control the regime dynamics - residence time, frequency and intensity - of the L84 model with appropriate nudges that are applied at critical times, using Local Lyapunov Exponents to trigger control, and a dynamic optimization that considers sequential nudges with bounded energy input, ensemble forecasts of the response with a randomized selection of the actual emergent state trajectory. Implications for real world systems are discussed.

We invite collaboration to develop Weather Jiu Jitsu as a community research platform.